

School of Computer Applications and Technology

Semester End Examination - Jan 2026

Masters of Computer Applications

Duration : 120 Minutes

Max Marks : 50

Sem I - E1PA103B - Problem Solving and Computer ProgrammingGeneral Instructions*Answer to the specific question asked**Draw neat, labelled diagrams wherever necessary**Approved data hand books are allowed subject to verification by the Invigilator*

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| 1) | List any four format specifiers in C with one example each. | CO1(5) |
| 2) | Write the steps involved in the compilation and execution process of a C program. | CO2(5) |
| 3) | Explain arrays as pointers. Provide an example where an array name acts as a pointer. | CO3(5) |
| 4) | Differentiate between Structure and Union with suitable example. | CO3(7) |
| 5) | Write a program to input elements in a 2D array (3×3), calculate sum of each row, and print results. | CO4(7) |
| 6) | Analyse the behavior of partial initialization. If you declare <code>int numbers[5] = {10, 20};</code> , what specific values are stored in <code>numbers[2]</code> , <code>numbers[3]</code> , and <code>numbers[4]</code> ? | CO4(7) |
| 7) | (a) What is a "Null Pointer"? What is the specific keyword used in C to initialize a pointer so it points to nothing? (b) Write a C program that declares an integer variable <code>num = 100</code> . Create a pointer <code>ptr</code> pointing to <code>num</code> . Use the pointer to increase the value by 50, then print the final value of <code>num</code> . | CO5(14) |

OR

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| (a) Explain the difference in search behavior between including a header file using angle brackets versus double quotes "file.h". (b) Identify the correct syntax to define a symbolic constant for PI. Explain why adding a semicolon (;) at the end of a #define statement is usually a syntax error when the macro is used. #define PI 3.14; vs #define PI 3.14 (c) Write the correct syntax to declare a pointer variable named <code>ptr</code> that points to a float variable. | CO6(14) |
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